

$$w^* \underset{\uparrow}{\text{argmin}} \sum_{i=1}^n \log(1 + \exp(\underbrace{y_i w^T x_i}_{\uparrow})) \rightarrow$$

$$\text{def } z = y_i w^T x_i$$

google $\rightarrow$ plot

$$= \underset{w}{\text{argmin}} \sum_{i=1}^n \log(1 + \exp(-z_i))$$

(+ve)

Best*

$$\exp(-z_i) > 0$$

$$\text{i.e. } \log(1 + \exp(-z_i)) > 0$$

model

↑
LNN

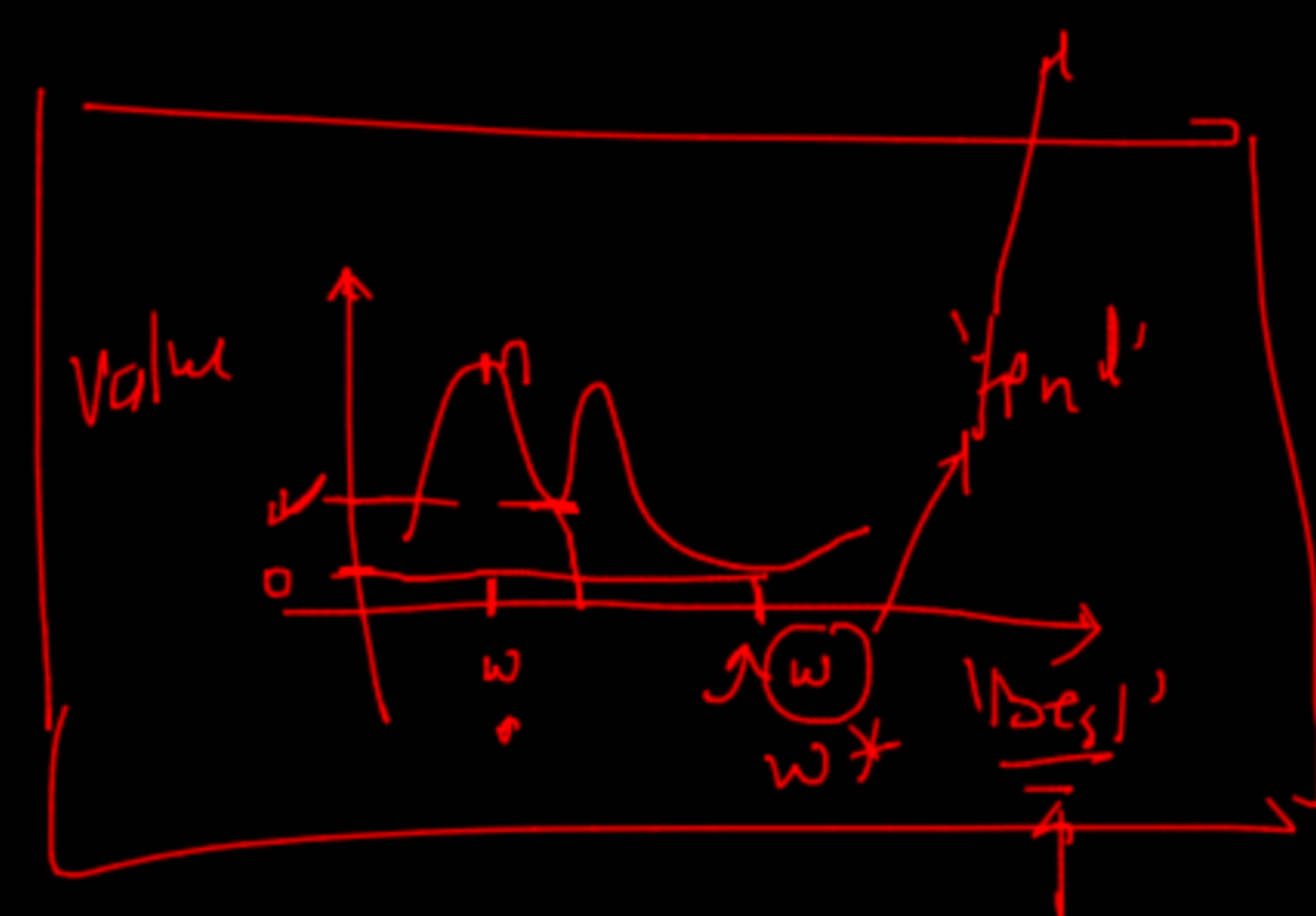
Hypen-params

↑
 $n = 1, 2, 3, \dots$

↑
overfit or underfit

BEST k

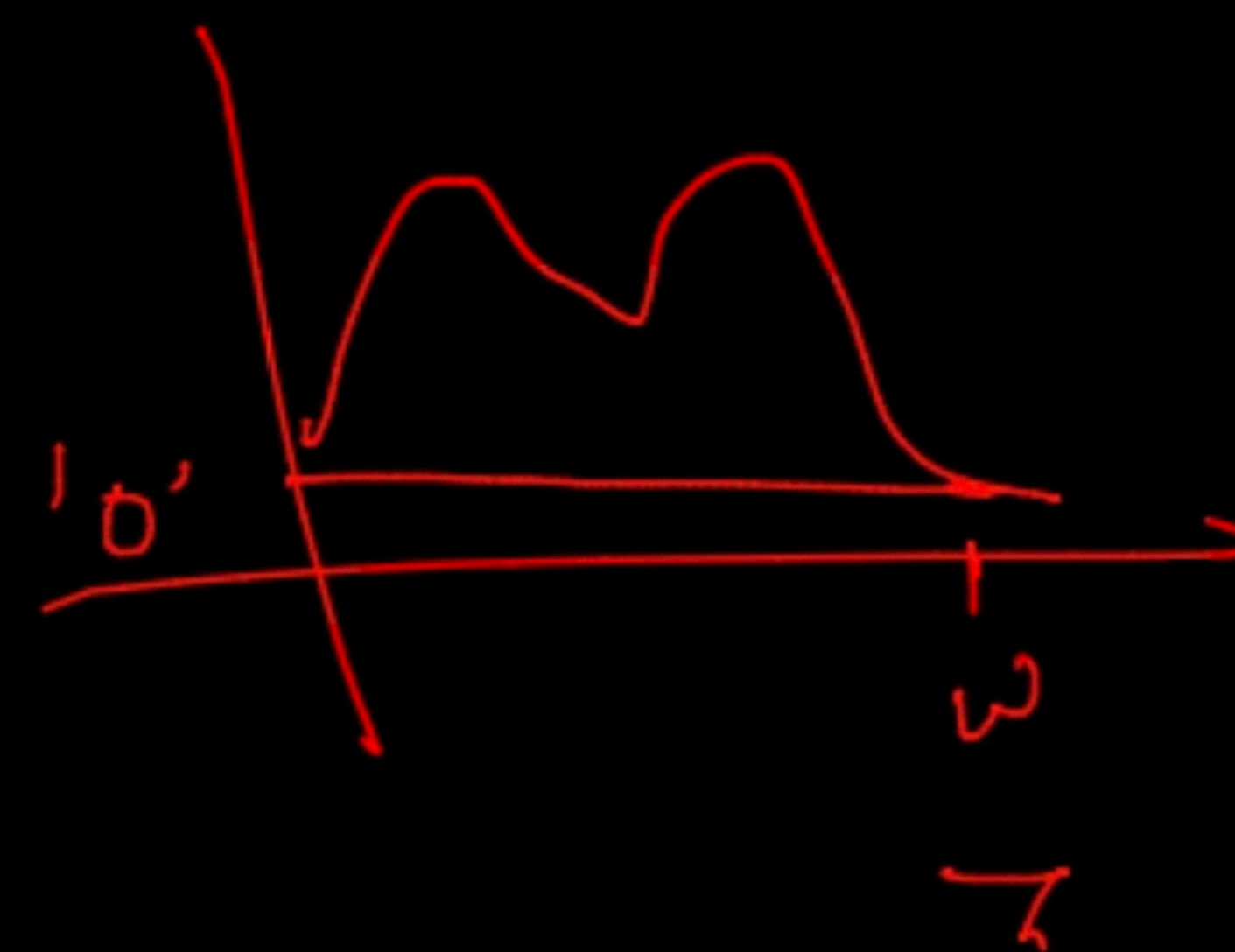
+ Best-fit



ie

$$\sum_{i=1}^n \log(1 + \exp(-z_i)) \geq 0$$

minimum value $\rightarrow 0$ when $10'$



$z_i \uparrow$

$\log(1)$

$z_i \rightarrow \infty$

$z_i \rightarrow \infty$

$\exp(-z_i) \rightarrow 0$

ie. $\log(1 + \exp(-z_i)) \rightarrow 0$

$$z_i = y_i \underset{\uparrow}{w}^T \underset{-}{x_i}$$

$$\mathcal{D} = \{ \langle x_i, y_i \rangle^i \}_{i=1}^n$$

$z_i \rightarrow \infty \rightarrow$ modify my (w) such way that $[z_i \rightarrow \infty]$

$$① Z_i = +ve$$

$Z_i \rightarrow$ incorrectly classified

$$② Z_i = +\infty \quad (\text{minimum})$$

$\rightarrow$ If I pick my (w) s.t.

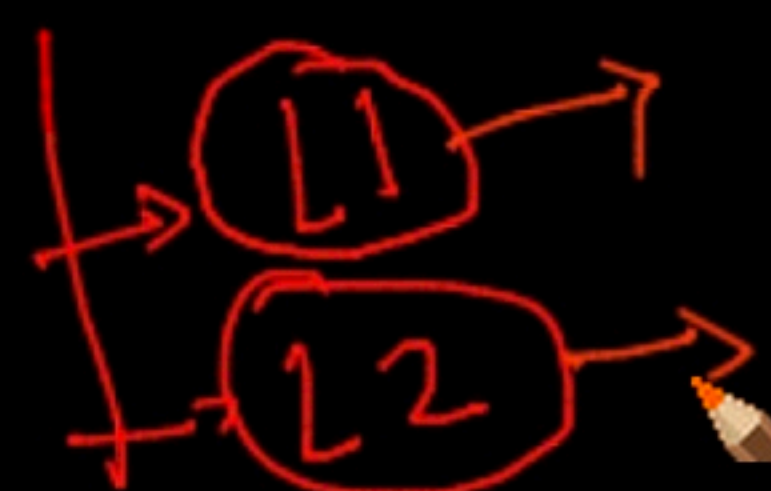
(a) all training points are classified correctly

$$① Z_i \rightarrow \infty$$

then $\boxed{w}$

'overfitting'

~~Regular~~ Regularization



$$w^* = \arg \min_w \underbrace{\sum_{i=1}^n \log(1 + \exp(-y_i w^T x_i))}_{\text{logistic loss for training data}} + \underbrace{\lambda \cdot w^T w}_{\text{L2 = regular}}$$

sklearn →
↑

w^* - argmin
↑

Hyper-parameters

[---] [---]

2.7

will avoid $w_i \rightarrow +\infty$
 $w_i \rightarrow -\infty$

KNN
↑
 $k=3$

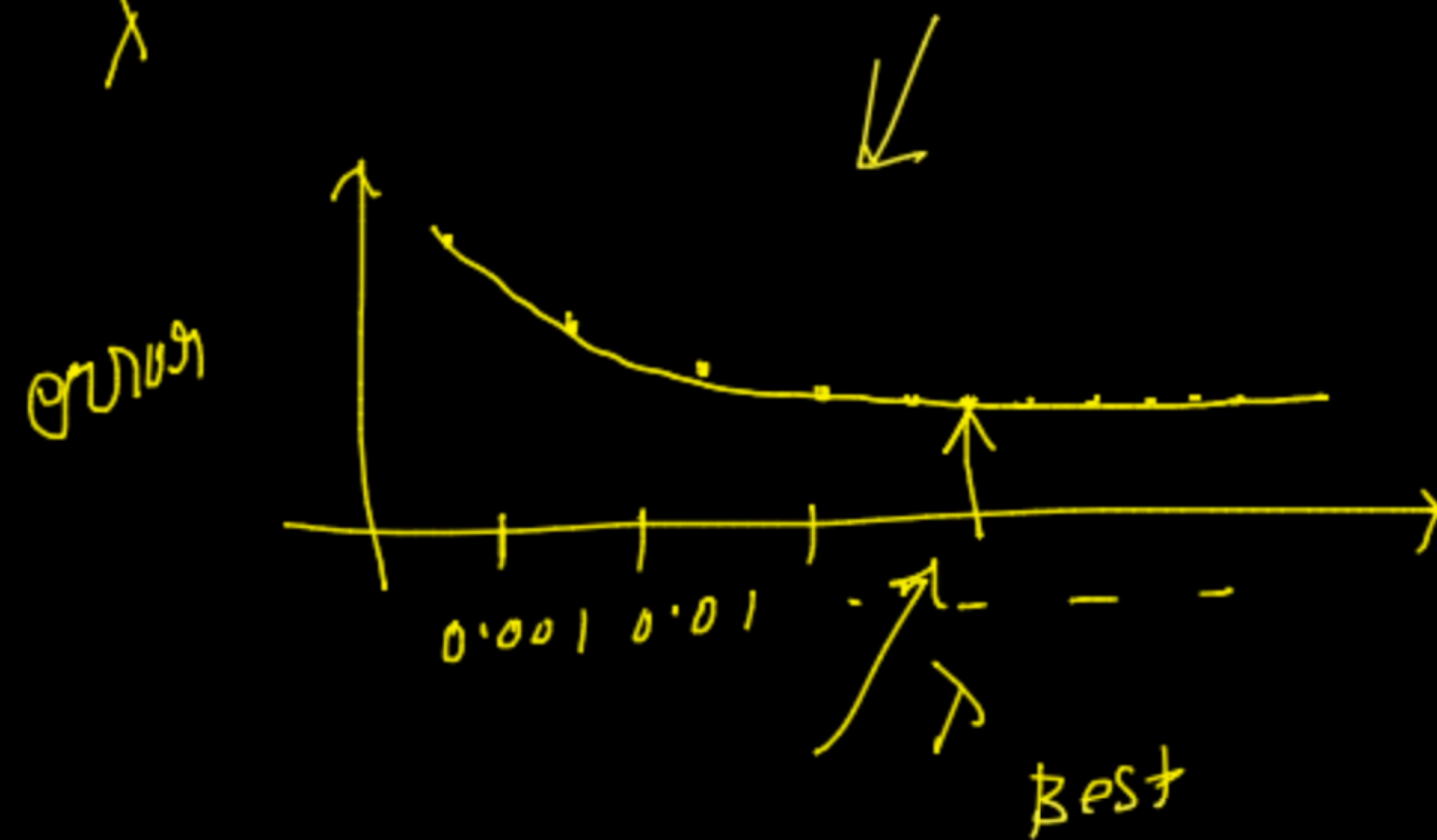
LR
↑
 $\left[\begin{array}{l} \lambda \\ \text{Reg} \rightarrow L1/L2? \end{array} \right]$

→ Hyper-parameter : search / optimisation

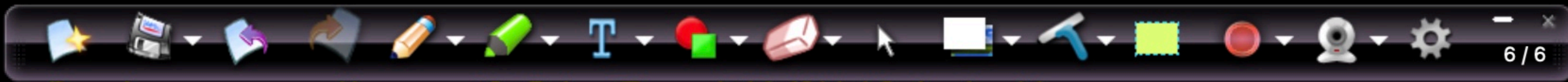
LR → λ
↑

$\hat{y}$
↑
 $\hookrightarrow \text{LR}(\lambda=0.01) \rightarrow$

$m = \text{LR}(\lambda=0.01, \text{penalty: } L1)$
 $m.\text{fit}(data)$
 $\text{Ave} \rightarrow \text{error} \rightarrow 0.54$



$m = \text{LR}(\lambda=0.001, p=L2)$
 $\hookrightarrow 0.56$



Gridsearch CV



OK

Random search CV



parameter

$$Solu = [nbf, n_{\text{neuro}}]$$

$$\lambda = [0.001, 0.01, 0.1]$$

$$penalty = [L1, L2]$$

$$mk$$

$m_1 \rightarrow 6 \text{ model}$

$k - \text{Hyper-parameters}$

Annotations: 'nbf' is circled in the solution vector. '0.001' is circled in the lambda vector. 'L1' is circled in the penalty vector. Arrows point from 'parameter' to the lambda and penalty vectors. An arrow points from 'mk' to the 'Hyper-parameters' text.